Community Security Alert System (CSAS): A Systems Analysis of Campus Safety as a Socio-Technical System

Workshop No. 1 — Revised Submission

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***Abstract*—This report presents a systems analysis of campus safety understood as a socio-technical system whose emergent property—community security—arises from the interaction of three coupled subsystems: incident generation, awareness propagation, and institutional response. Following the leverage-point framework of Meadows and the soft-systems methodology of Checkland, the analysis first characterizes the system under study, then identifies the structural disconnects that bound its current performance, and only then evaluates candidate interventions. Primary evidence was collected through structured interviews with security staff, a student survey ($n = 20$), direct observation across four campus zones (8 sessions), institutional document analysis, and benchmarking of three existing alert platforms in Bogotá. The data converge on a single structural finding: the system is bounded not by the willingness of its members to act, but by the latency, reliability, and traceability of the information flow connecting incident, awareness, and response. A causal loop analysis reveals a reinforcing under-reporting loop driving normalization of risk, and a balancing alert-fatigue loop constraining naive amplification of awareness. Four candidate interventions are evaluated against six criteria; the Community Security Alert System (CSAS) is recommended as the intervention with the highest leverage on the information-flow bottleneck, conditional on the explicit validation of three assumptions and the management of four classes of unintended consequences identified in the analysis.**

***Index Terms*—systems analysis, socio-technical systems, leverage points, causal loop diagrams, feedback dynamics, campus safety, crowd-sourced incident reporting, trade-off analysis, alert fatigue**

I. System Characterization

A. The Campus Security System as Phenomenon Under Study

University campuses operate as complex socio-technical systems in which thousands of students, faculty, staff, and visitors interact daily across spatially distributed zones with heterogeneous risk profiles. Within this system, the emergent property of community safety arises from the interaction of three interlocking subsystems: (i) an *incident-generation* subsystem, driven by spatial, temporal, environmental, and behavioral factors; (ii) an *awareness* subsystem, through which information about events propagates among community members; and (iii) a *response* subsystem, through which institutional actors coordinate intervention once an incident becomes known.

Empirical evidence collected for this study reveals a structural disconnect between these three subsystems. Incidents are generated at rates substantially higher than those captured by institutional records: 35% of surveyed students reported direct experience of a security incident, against significantly lower figures in formal logs (see Sections II and VIII). Awareness propagates slowly and unevenly through informal channels with no shared memory. Response is initiated only after an incident reaches the security team through high-friction paths—primarily phone calls or physical approach—introducing reporting latencies that interview data estimate at several minutes per event. The system's effective security performance is therefore bounded not by the willingness of its members to act, but by the latency, reliability, and traceability of the information flow that connects incident, awareness, and response.

The Community Security Alert System (CSAS) is proposed in this report as a targeted intervention on that information flow. Following the leverage-point framework of Meadows [3], CSAS does not aim to add new sensing or enforcement capabilities to the campus; rather, it aims to modify a single high-leverage variable—the friction of community-to-institution communication—while preserving the existing institutional response infrastructure (campus security team, SIURE, local authorities). **CSAS is therefore not the system under analysis in this document but a candidate intervention** whose justification depends on a prior, system-level characterization of the campus security environment. The remainder of this report develops that characterization and, on its basis, evaluates whether CSAS constitutes the appropriate response.

B. System Boundaries

Defining clear boundaries is essential for scoping the analysis and preventing scope overextension. Table I separates what falls inside the system under study from what constitutes its external environment. The boundary is drawn around the *phenomenon* (campus safety) rather than around the proposed intervention (CSAS); this distinction is what allows the analysis to remain valid independently of the specific technological solution ultimately adopted.

TABLE I: System Boundary Definition — Campus Safety as System Under Study

Inside the system (scope)	**Outside the system (environment)**
Incident-generation dynamics on campus	City-level crime patterns
Community awareness and reporting behavior	Public telecommunications and ISP networks
Institutional verification and response workflow	Physical campus infrastructure (gates, fences)
Information flows among students, staff, and security personnel	Local law enforcement (Police, EMS) procedures
Existing reporting channels (SIURE, phone, in-person)	Campus surveillance hardware (CCTV)
Incident memory and historical records	Non-registered external visitors

C. Stakeholder Network

Six stakeholder groups participate in the campus safety system. Their characterization here emphasizes *role within the system* (what each group contributes and consumes), not their relation to any particular technology. Table II formalizes this mapping; Section IV-B extends it into a relational view.

D. System Purpose: Closing Operational Gaps

The purpose of any intervention on the campus safety system is best expressed not as a list of features to be built but as a set of measurable changes in the system's behavior. Drawing on the gap analysis quantified in Section VIII-C (Table XII), the system-level purposes that justify intervention are:

1) Reduce the average incident-to-response latency from $\sim$11.4 min to under 5 min.
2) Raise the share of incidents captured in institutional records from $\sim$41% to at least 85%.
3) Raise sustained community participation in safety reporting from $\sim$12% to at least 60%.
4) Raise the alert-delivery success rate from $\sim$87% to at least 99% on critical alerts.

Each of these statements describes a change in a state variable of the campus safety system, not a feature of CSAS. Any intervention—CSAS or otherwise—will be judged by its observable contribution to these state changes (see Section VII-D).

E. Operational Environment

The system operates within four interacting dimensions of its environment. The *physical* dimension encompasses the campus itself: academic buildings, parking lots, residences, and the adjacent public thoroughfares that constitute commuting routes; this dimension determines spatial risk distribution. The *social* dimension comprises the heterogeneous community of students, professors, staff, and visitors who simultaneously generate and consume safety information; this dimension determines reporting behavior and trust. The *institutional* dimension covers the coordination between university administration, campus security, and local law-enforcement agencies; this dimension determines response capacity. The *infrastructural* dimension—which includes mobile connectivity, the institutional SIURE channel, existing surveillance hardware, and the absence of any centralized mobile reporting tool—constrains what kinds of intervention are feasible. The infrastructural dimension is the only one in which CSAS would introduce new components; the other three are inherited from the operational reality of the campus and must be accepted as boundary conditions of the analysis.

F. System Context Diagram

Fig. 1 presents the system context diagram, showing the information flows that exist among external actors of the campus safety system. The diagram intentionally depicts the system in its current (pre-intervention) state, so that the gaps it reveals justify the design of any subsequent intervention. The information flow from students/staff to the security team is informal, low-bandwidth, and unrecorded; the flow from the security team to local authorities is selective and post-hoc; and the upward flow to university management is sparse and aggregated.

Fig. 1 from the original submission is reused here without modification, as it correctly characterizes the pre-intervention information flow among external actors. Source file: `fig_context.pdf`.

Fig. 1: System context diagram of the campus safety system. Arrows indicate the direction of information flow between each stakeholder group. Source: own elaboration (2026).

G. Proposed Intervention: The CSAS

Only after the system has been characterized is it productive to describe the proposed intervention. The Community Security Alert System (CSAS) is a collaborative socio-technical artifact whose specific contribution to the system is to lower the friction of community-to-institution incident reporting and to add structured memory to the campus safety system. Architecturally, it is composed of an interface layer (mobile, web, push channel), a processing core (REST API, verification logic, geolocation, authentication, alert dispatch), and a data layer (incidents database, media store, analytics, audit logs). Fig. 2 presents this architecture, which is reused from the original submission. The critical point is that the architecture is a *consequence* of the analytical findings in subsequent sections, not a starting assumption.

TABLE II: Stakeholder Roles in the Campus Safety System (Pre-Intervention)

Stakeholder	System role	Information they generate	Information they need
Students	Primary sensors and primary subjects of risk	First-hand observation of incidents and suspicious activity	Zone-specific risk awareness in near real time
Professors and administrative staff	Secondary sensors; institutional bridges	Reports of disruptions affecting academic activity	Safety status of buildings and routes
Campus security team	Operational responders; verification authority	Verified incident records, response outcomes	Aggregated and prioritized incoming reports
University management	Strategic decision-makers	Resource allocation, policy directives	Trends and analytics for planning and accountability
Local authorities (Police, EMS)	External responders for escalated cases	Outcome of external intervention	Verified, high-confidence incident hand-offs
System administrators	Custodians of technical operation	Operational metrics and audit data	Platform health, access patterns, security events

Fig. 2 from the original submission is reused here. Source file: `fig_architecture.pdf`.

Fig. 2: Architecture of the proposed CSAS intervention. Node size reflects relative centrality within the artifact; the REST API is the highest-connectivity node. The artifact is presented here as a consequence of the system analysis, not as its starting point. Source: own elaboration (2026).

II. Data Collection Plan and Execution

A. Data Collection Plan

To characterize the campus safety system, we designed a data collection plan combining six methods. The plan was guided by two principles: *methodological triangulation*, so that no finding would rest on a single source, and *cross-sectional coverage*, so that perspectives from students, security staff, and direct field observation would be represented in the evidence base. Table III summarizes the six methods.

B. Interviews

We interviewed five security staff members and two administrative staff members. The objective was to characterize the institutional response subsystem from the inside. Three patterns recurred across nearly all interviews. First, the latency from incident occurrence to staff notification is high and variable, because notification depends on a community member placing a phone call or physically approaching a guard. Second, there is no centralized memory: each incident is handled in isolation and the institution loses the ability to identify spatial or temporal patterns. Third, staff articulated a concrete need for notifications that include verified location data, not verbal descriptions. These three findings are not features demanded for CSAS—they are structural deficiencies of the current information flow.

C. Student Survey

We distributed an anonymous online survey to 20 students measuring perceived safety, incidence experience, and willingness to adopt a mobile reporting channel. Table IV and Fig. 3 present the raw data and the headline finding.

Fig. 3 from the original submission is reused here. Source file: `fig_pie.pdf`.

Fig. 3: Proportion of surveyed students who experienced or witnessed a security incident ($n = 20$). 35% reported a positive answer. Source: own elaboration (2026).

Three patterns emerge from the dataset and are interpreted here in system terms rather than as user preferences:

- Reported incidents (theft, harassment, suspicious activity) concentrate in afternoon and night hours, confirming that the incident-generation subsystem is non-stationary in time.
- 100% of students who experienced an incident expressed willingness to adopt a reporting channel, versus 69% among those who did not. This is consistent with a Bayesian update of perceived utility upon exposure to the underlying risk—a behavioral pattern relevant to adoption dynamics.
- Several students reported feeling safe while simultaneously reporting having experienced an incident. This is a candidate signature of risk normalization, an emergent property of the social subsystem worth investigating in subsequent phases.

D. Direct Observation

Eight observation sessions were carried out across four campus zones: the main entrance, the library area, the parking lot, and the student residence area. Each zone was observed in at least two distinct time slots. Table V summarizes the record.

Four findings emerge. The parking lot at night is the highest-risk observed zone, with poor lighting and low foot traffic as contributing factors. The student residence area exhibits two specific points with deficient lighting and no recorded incident—a latent risk invisible to official records. The two suspicious events recorded by our team were not reported by any other witness during the observation period, which independently corroborates the under-reporting hypothesis. Finally, both suspicious events coincided with rainy night conditions, suggesting a possible coupling between the incident-generation subsystem and environmental variables.

TABLE III: Data Collection Plan — Six Methods Applied to Characterize the Campus Safety System

Method	Instrument	Sample	Duration	Objective
Interviews	Structured guide	5 security staff, 2 administrative staff	1 week	Characterize the existing reporting workflow and identify the operational pain points perceived from inside the institutional response subsystem.
Surveys	Online form	20 students	2 weeks	Measure perceived safety, incidence rate, and willingness to adopt a mobile reporting channel within the community.
Direct observation	Observation record	4 campus zones, 8 sessions	1 week	Capture spatial and temporal patterns of activity and suspicious behavior independently of self-reported data.
Document analysis	Institutional reports	Existing campus incident records	3 days	Establish a baseline of officially recorded incidents and compare it against field findings to test the under-reporting hypothesis.
Benchmarking	Comparative analysis	3 alert platforms in Bogotá	1 week	Identify proven design patterns and known failure modes in adjacent systems, to constrain the space of feasible interventions.
Weather monitoring	Public weather data	Teusaquillo locality	1 week	Explore whether environmental conditions correlate with the activity and incident patterns observed in the field.

TABLE IV: Raw Survey Data ($n = 20$). Variables: safety perception, incident experience, incident type, willingness to use an alert channel, and time of day.

Table III from the original submission is reused here. Source file: `tbl_survey.pdf` *or in-document re-typesetting.*

TABLE V: Direct Observation Record — 8 Sessions Across 4 Campus Zones.

Table IV from the original submission is reused here. Source file: `tbl_observation.pdf` *or in-document re-typesetting.*

E. Document Analysis

Available institutional security reports were reviewed for historical patterns. The most salient observation is the magnitude of the gap between formally recorded incidents and the prevalence implied by survey and observation data. Theft and unauthorized access are the two most documented categories, consistent with what students reported in the survey.

F. Benchmarking

Three existing alert platforms operating in Bogotá were analyzed. Table VI compares them and extracts the design implications relevant to the analysis of candidate interventions in Section VI.

G. Weather Monitoring

Weather conditions in the Teusaquillo locality were tracked for one week and cross-referenced with the observation log. Both suspicious events recorded during the observation phase coincided with rainy night conditions. We classify this finding as *hypothesis-grade* rather than conclusive, given the limited observation window. Its implications are discussed in Section VIII.

III. Data Quality Evaluation

A. Identified Limitations

A rigorous analysis must explicitly bound the reach of its conclusions. Table VII lists, for each method, the most material limitation and its implication for the strength of the findings derived from it.

B. Why the Data Remains Useful

Three reasons support the use of this evidence base for the present phase of the analysis. First, the three most consequential findings—spatio-temporal concentration of incidents, low reporting culture, and preference for a mobile channel—appear independently in the interviews, the survey, and the direct observation. The convergence of independent methods on the same structural conclusion strengthens confidence in that conclusion beyond what any single method could provide. Second, all data is primary and collected first-hand by the team, eliminating intermediary distortion and preserving traceability. Third, the evidence is sufficient to support *system-level* conclusions, even if it is insufficient to support fine-grained quantitative claims. The limitations identified above do not invalidate the present analysis; they define the validation agenda for the next phase.

IV. System Elements, Relationships, and Causal Structure

A. Identification of System Elements (Pre-Intervention)

The campus safety system, prior to any technological intervention, comprises seven interacting elements. This decomposition deliberately excludes the components of CSAS itself, in order to characterize the system on terms that are independent of the proposed solution:

1) **Community members** (students, faculty, staff, visitors) as both potential subjects of risk and potential observers.
2) **Risk events**: thefts, harassment, suspicious activity, medical and accidental emergencies.

TABLE VI: Benchmarking of Existing Security Alert Platforms in Bogotá

Platform	Strengths	Weaknesses	Implication for intervention design
Policía Nacional / Vigila App	Direct integration with police; national coverage	Slow response; depends on user placing a phone call	Authority integration is a useful design pattern, but integration alone does not reduce latency.
Neighborhood WhatsApp networks	High adoption; familiar channel; instant propagation	No verification, no records, panic-prone	Immediacy is desirable, but unverified amplification triggers the alert-fatigue balancing loop (Section V-B).
SIURE UD (internal system)	Already exists; staff are trained on it	Email and phone only; no mobile, no real-time capability	Any new intervention should complement, not replace, the existing institutional process.

TABLE VII: Limitations by Data Collection Method and Their Implications

Method	Limitation	Implication
Survey	Sample of 20 students	Findings are directional indicators, not statistically representative prevalence estimates.
Survey	Self-reported, unverified responses	Risk of social-desirability bias; partially mitigated by anonymous administration.
Observation	Only 4 zones, 8 sessions	Indoor corridors, cafeterias, and transit zones between buildings were not covered.
Observation	Possible observer effect	Behaviors may have shifted in the presence of note-taking observers; mitigation through unobtrusive positioning.
Document analysis	Limited access to disaggregated data	No zone- or time-specific breakdown of historical incidents was available, preventing finer historical validation.
Benchmarking	Public information only	No access to real usage data or internal metrics of compared platforms.
Weather monitoring	One-week window	Insufficient to establish correlation; findings are hypothesis-grade only.

3) **The physical environment**: campus zones differentiated by lighting, foot-traffic density, time-of-day exposure, and weather conditions.
4) **Existing reporting channels**: phone calls, physical approach to guards, the SIURE institutional channel.
5) **Campus security personnel** and their existing workflow for verification and dispatch.
6) **Institutional memory**: the (currently sparse and non-systematic) record of past incidents.
7) **Local authorities** (Police, EMS) as external responders for escalated cases.

It is the *relationships* among these elements—in particular, the bandwidth and latency of information flow between elements 1 and 5 through element 4—that determine system performance.

B. Stakeholder Interaction Diagram

Fig. 4 presents the stakeholder interaction diagram, which complements the context diagram of Fig. 1 by depicting pair-wise relationships, the nature of each link, and the asymmetries among them.

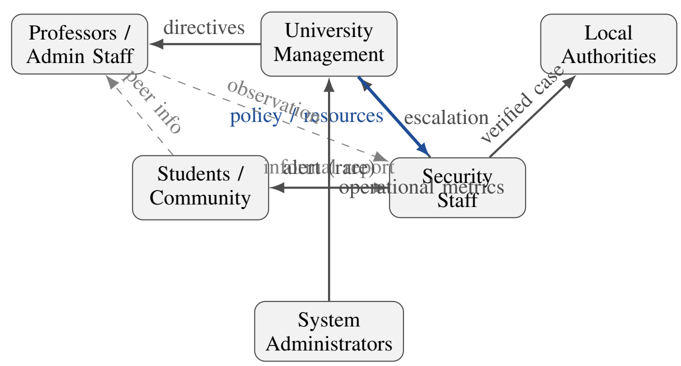

Fig. 4: Stakeholder interaction diagram of the campus safety system. Solid arrows denote structured, recorded information flows; dashed arrows denote informal, unrecorded, or low-bandwidth flows; thick blue arrows denote relations of authority and resource allocation. The asymmetry is the central finding: information flowing *into* the security staff is informal and lossy, whereas information flowing *out of* the security staff toward the community is structurally underdeveloped. Source: own elaboration (2026).

C. Cause-Effect Analysis of Under-Reporting

The single most consequential structural deficiency identified in the data is the gap between actual incidence and recorded incidence. Fig. 5 formalizes this in an Ishikawa (fishbone) diagram, mapping the contributing causes across four conventional categories.

The diagnostic value of Fig. 5 is that it constrains the design space of feasible interventions: any intervention that addresses only one branch (for example, a purely educational campaign addressing the People branch) is unlikely to close the under-reporting gap on its own. This is one of the explicit justifications for preferring a multi-pronged intervention in Section VI.

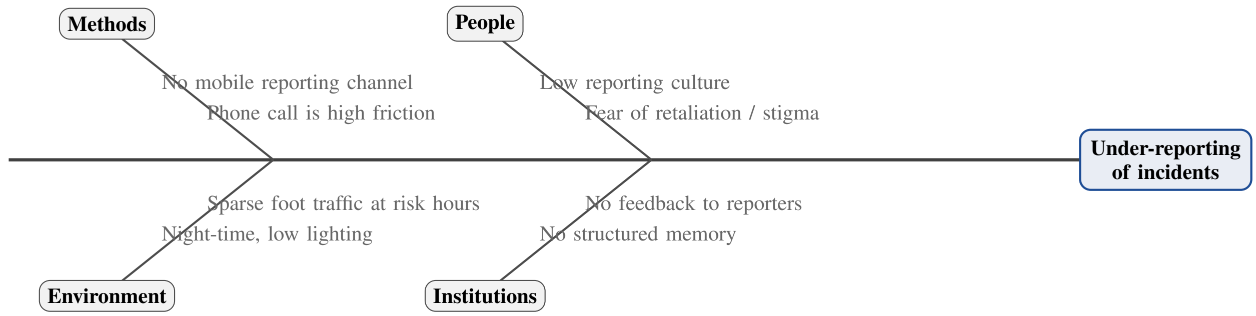

Fig. 5: Ishikawa diagram of under-reporting. Each branch maps a class of root causes contributing to the observed gap between actual and recorded incidence. The diagram makes explicit that under-reporting is not attributable to a single failure but to the convergence of methodological, behavioral, environmental, and institutional factors. Source: own elaboration (2026).

D. *Information Flow Diagram: Current vs. Target State*

Fig. 6 presents a side-by-side comparison of the current information flow (left) and the target information flow under the proposed intervention (right). The visualization makes the leverage point explicit: the intervention compresses the chain from event to response, adds structured memory, and introduces a feedback channel to the community.

V. System Dynamics: Sensitivity, Feedback Loops, and Emergent Behavior

A. *Causal Loop Diagram*

The static element-relationship view of the previous section is necessary but insufficient: the campus safety system also exhibits *dynamic* behavior driven by feedback loops. Fig. 7 presents the causal loop diagram (CLD) of the system. Each arrow is labeled with a polarity: a positive sign (+) indicates that the variables move in the same direction, a negative sign ($-$) indicates that they move in opposite directions. Closed paths form either reinforcing loops (denoted R) or balancing loops (denoted B).

B. *Reinforcing and Balancing Loops*

Three loops dominate the dynamics of the system, and an effective intervention must reason about all three simultaneously.

a) Loop R1 — Virtuous reinforcement: Reports submitted increase community awareness, which increases personal vigilance, which reduces risk events, which reduces the input to the loop. Inverted, the same loop produces a vicious cycle: fewer reports lead to lower awareness, which leads to lower vigilance, which permits more incidents but fewer reports of them. The system sits today on the vicious side of R1; activating its virtuous mode is the principal goal of any intervention.

b) Loop R2 — Under-reporting trap: The current state of the system also sustains a self-reinforcing under-reporting trap: low reporting yields low awareness, which fosters risk normalization, which increases the perceived friction of reporting ("why bother if nothing happens?"), which further suppresses reports. R2 is consistent with the survey finding that several students who reported having experienced an incident also reported feeling safe.

c) Loop B1 — Alert fatigue: The balancing counterweight to any awareness-increasing intervention is alert fatigue: as the volume of alerts grows, the marginal attention paid to each one falls, and beyond a threshold the intervention loses signal value. B1 is the system-level reason why verification matters: without a verification stage, the intervention amplifies B1 faster than it activates R1, and net effectiveness declines. The verification component of CSAS is thus not a feature chosen for engineering elegance but a structural requirement imposed by the dynamics of the system.

C. *Parameter Sensitivity Analysis*

The behavior of the system is dominated by a small set of operational parameters. Table VIII formalizes the analysis by linking each parameter to its observable effect, the supporting evidence, and the loop on which it acts.

D. *Non-Linear and Emergent Effects*

Two non-linearities deserve explicit attention. First, the relationship between active-user count and simultaneous report volume is non-linear: a small increase in active users can produce a disproportionately large spike in concurrent reports during high-activity periods, which stresses the verification stage and may trigger B1. Second, spatial and temporal clustering means that alert activity is not uniformly distributed across the campus or the day; the system therefore requires zone-weighted and time-weighted priority logic rather than uniform processing of incoming reports. Both non-linearities are properties of the campus safety system itself, not of any particular technological artifact, and they constrain the design of any intervention.

VI. Alternative Interventions and Trade-Off Analysis

A. *Candidate Interventions*

Following the system-first principle, the analysis evaluated four candidate interventions on the campus safety system before settling on a recommendation:

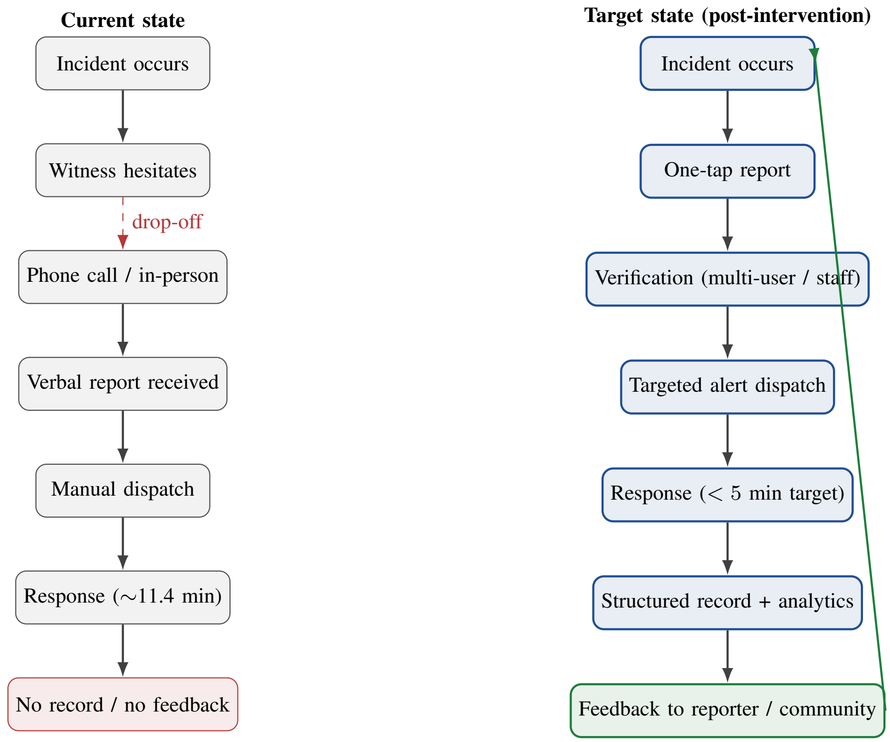

Fig. 6: Information flow in the campus safety system: current state (left) and target state under intervention (right). Dashed red arrows indicate flows with substantial drop-off. The green arrow on the right closes the feedback loop, transforming the linear current process into a learning loop. Source: own elaboration (2026).

1) **A1 — Physical-security expansion**: more guards, more CCTV coverage, improved lighting in identified risk zones.
2) **A2 — Awareness and training programs**: mandatory reporting workshops, posters, induction sessions to address the reporting-culture branch of Fig. 5.
3) **A3 — Direct integration with city emergency services** (e.g., a 123/911 fast-track from campus).
4) **A4 — Crowd-sourced mobile reporting platform with verification (CSAS, our proposal)**.

These four alternatives do not exhaust the design space, but they span the principal dimensions on which interventions can vary (infrastructure-heavy vs. behavior-focused vs. external-coordination vs. information-flow-focused).

B. Comparative Evaluation

Table IX compares the four candidates on six criteria derived from the operational gaps quantified in Section VIII-C.

C. Justification of CSAS

CSAS is recommended because it is the only candidate that acts directly on the structural variable identified as the system's binding constraint: the friction of community-to-institution information flow. A1 raises baseline security but leaves the under-reporting trap untouched. A2 addresses one branch of the fishbone but leaves the methods and institutional branches unchanged; the data do not support the hypothesis that awareness alone closes the gap. A3 introduces external coordination latency and does not solve internal reporting. A4—CSAS—combines mobile-channel design (methods branch), structured memory (institutional branch), and verification logic (B1 mitigation) in a single intervention. Crucially, the recommendation is conditional on the management of the assumptions and unintended consequences discussed in Section VII; if those assumptions fail, A4 reduces to A2 plus a software artifact and loses its leverage advantage.

VII. Risks, Assumptions, and Unintended Consequences

A. Explicit Assumptions and Validation Status

The analysis rests on five assumptions whose validation status is made explicit in Table X. Each row identifies the assumption, the evidence (if any) currently supporting it, the

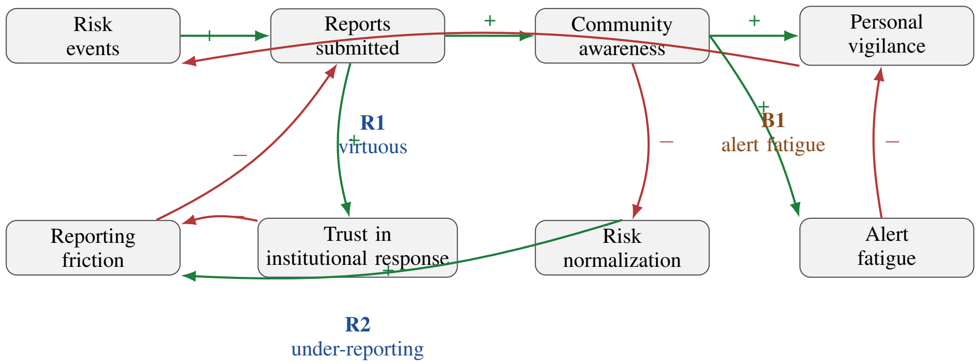

Fig. 7: Causal loop diagram of the campus safety system. Solid green arrows indicate positive polarity; solid red arrows indicate negative polarity. Loop R1 is the virtuous reinforcing loop that an effective intervention seeks to activate; loop R2 is the vicious reinforcing loop that sustains the current under-reporting state; loop B1 is the balancing loop that constrains naive amplification of awareness. Source: own elaboration (2026).

TABLE VIII: Parameter Sensitivity Analysis with Loop Attribution

Parameter	Possible change	System effect	Supporting evidence	Acts on
Number of reports	Sharp increase	Verification capacity is exceeded, latency rises, B1 dominates	35% survey incidence suggests latent volume	R1, B1
User participation	Low engagement	Real risks remain invisible, R2 persists	Low reporting culture across all methods	R1, R2
Time of day	Night-time rise	Incident frequency rises non-linearly	All suspicious events observed in evening/night	R1
Location	High-risk zones	Localized alert clustering	Parking lot consistently identified as highest-risk	R1
Weather conditions	Adverse weather	Visibility drops, movement patterns shift	Both observed suspicious events on rainy nights (hypothesis-grade)	R1
Verification latency	Tighter threshold	Faster alerts at cost of more false positives, accelerating B1	Trade-off articulated in benchmarking	B1

TABLE IX: Trade-Off Comparison of Candidate Interventions

Criterion	A1: Physical security	A2: Awareness programs	A3: City emergency integration	A4: CSAS (proposed)
Leverage on root cause (info-flow friction)	Low	Medium	Low	High
Capital cost	High	Low	Low	Medium
Time to deploy	Long (months–years)	Short–medium	Long (procedural)	Short–medium
Dependence on external actors	Medium	Low	High	Medium
Effect on under-reporting (R2)	Indirect	Direct but slow	None	Direct and structural
Risk of triggering B1 (alert fatigue)	Low	Low	Low	Medium (mitigated by verification)

classification of its validation status, and the action proposed to validate it in the next phase.

B. Risk Register

Five risks materially affect the intervention. Table XI characterizes each by likelihood, impact, and proposed mitigation.

C. Potential Unintended Consequences

A rigorous systems analysis must anticipate that an intervention designed to improve a system will also produce effects that are not part of its design intent. We identify four classes of unintended consequences:

a) Vigilantism risk: A well-functioning awareness channel may encourage community members to act on reported incidents before institutional responders arrive. This is undesirable and must be mitigated by the design of the alert content itself (advisory rather than directive language) and by clear behavioral guidelines bundled with the platform.

b) Risk profiling and discrimination: Aggregated incident data could be used—intentionally or not—to profile specific zones, demographics, or individuals. The analytics layer of any implementation must therefore include explicit policies on aggregation, anonymization, and access control,

TABLE X: Explicit Assumptions, Evidence Base, and Validation Plan

#	Assumption	Supporting evidence	Status	Validation action (next phase)
A1	Community members will report if friction is reduced	Survey: 100% of incident-experienced respondents willing to adopt; benchmarking: high adoption of WhatsApp channels	Plausible	Pilot test with limited cohort, measure actual reporting rate vs. willingness
A2	Verification can be made fast enough for genuine emergencies	Benchmarking: "slow" is the principal weakness of existing platforms; design space analysis	Untested	Define explicit latency budget, measure under load
A3	Mobile connectivity is reliable across all campus zones	Not assessed in current observation phase	Unvalidated	Signal-coverage survey across the four observed zones
A4	Verification logic can keep false-positive rate below the B1 threshold	Theoretical: verification narrows the alert stream; no empirical measurement yet	Untested	Define numeric false-positive threshold; simulate or pilot
A5	The campus institutional infrastructure (SIURE, security team) will integrate rather than resist	Interviews: staff articulated a clear need; no formal institutional commitment yet	Partially supported	Stakeholder agreement and integration plan with university administration

TABLE XI: Risk Register for the Proposed Intervention

#	Risk	Likelihood	Impact	Mitigation
R1	Low sustained adoption ($<$ 60% target)	Medium	High	Institutional endorsement, induction integration
R2	Verification bottleneck under spike loads	Medium	High	Asynchronous verification queue, staff capacity planning
R3	Connectivity dead-zones break the alert chain	Medium	Medium	SMS fallback channel; signal-coverage survey
R4	False positives erode trust (B1 dominates)	Medium	High	Multi-user confirmation, staff moderation, reputation logic
R5	Misuse: harassment reporting weaponized, data privacy breaches	Low–Medium	High	Reporter anonymity, content moderation, compliance with Ley 1581/2012 [6]

beyond minimum compliance with Ley 1581/2012 [6].

c) Reduced personal vigilance: A counterintuitive consequence of an effective alert system is that community members may delegate vigilance to the platform ("the app will warn me"), reducing the baseline situational awareness on which the system itself depends. This is a Goodhart-style risk and is particularly difficult to detect by quantitative monitoring alone.

d) Fear amplification: An accurate alert system makes the system's incident reality visible. For communities accustomed to low visibility, the initial perception may be that safety has *worsened*, even though only the measurement has improved. Management of this perception is a non-technical implementation requirement.

D. Evaluation Criteria for the Proposed Intervention

The intervention will be evaluated against six criteria mapped to the operational gaps of Table XII:

1) **Median incident-to-response latency** (target: < 5 min).
2) **Incident record completeness** (target: $\geq 85\%$).
3) **Sustained community adoption** (target: $\geq 60\%$ of eligible community using the channel at least monthly).
4) **Alert delivery success rate** (target: $\geq$ 99% for verified critical alerts).
5) **False-positive rate on verified alerts** (target: below a threshold to be defined empirically based on B1 dynamics).
6) **Stakeholder satisfaction** (qualitative; campus security team, students, and university management surveyed periodically).

These criteria are deliberately stated as observable system-level metrics, not as features of the artifact.

VIII. Critical Assessment

A. Strengths of the Analysis

Several elements of this study are methodologically robust. The use of six complementary data collection methods supports triangulation: the three structural findings (underreporting, spatio-temporal concentration, absence of a centralized channel) appear independently in interviews, surveys, and direct observation, which provides mutual validation. All data is primary and collected first-hand, ensuring traceability. The boundary definition (Section I-B) is drawn around the phenomenon (campus safety) rather than around the proposed artifact (CSAS), which keeps the analysis valid independently of the specific solution adopted. The sensitivity analysis goes beyond description to identify which parameters dominate system behavior, which is essential for prioritization in design.

B. Limitations

The reach of the conclusions is bounded by the limitations summarized in Table VII. The survey sample of 20 students cannot support statistical generalization. The observation phase covered only four zones in eight sessions, leaving indoor spaces and transit zones uncharacterized. The institutional document analysis was limited by restricted access to disaggregated data. The weather-correlation finding is hypothesis-grade only and would require a longer observation window and more events to support a causal claim. These limitations define the validation agenda for the next phase; they do not invalidate the system-level conclusions.

C. Operational Gap Quantification

Table XII quantifies the operational gap between the current state of the campus safety system and the target state that the intervention seeks to achieve. The gaps are not merely technical; they reflect deeply embedded behavioral and institutional patterns, which is why some of them (notably the adoption gap) cannot be closed by the artifact alone but require parallel institutional action.

IX. Conclusions

This analysis approached campus safety as a socio-technical system under study, not as a problem to be solved by a predetermined technological artifact. Four conclusions follow from that approach, each formulated as an insight about the system rather than as a restatement of the proposed solution.

a) The system is information-bound, not capacity-bound: The dominant constraint on the campus safety system is the latency and reliability of the information flow connecting incident, awareness, and response. This conclusion is supported by the triangulation of interviews, survey, and observation, and it has a clear design implication: an intervention that adds enforcement capacity without addressing the information flow is unlikely to shift system performance.

b) The system sits in a self-reinforcing under-reporting trap: The causal loop analysis shows that the current state of the system is sustained by a vicious reinforcing loop (R2 in Fig. 7): low reporting yields low awareness, which fosters risk normalization, which further suppresses reporting. Breaking this loop requires acting on the friction of reporting, not on the willingness of individuals to report—the willingness already exists, as evidenced by the 100% adoption intent among incident-experienced respondents.

c) User participation and verification latency are the dominant parameters: The sensitivity analysis identifies user participation and verification latency as the two parameters with the largest influence on system behavior. The first activates the virtuous loop R1; the second prevents the balancing loop B1 from dominating. Any intervention that scores poorly on either parameter cannot succeed, regardless of its other features.

d) Verification is not a feature, it is a structural requirement: The presence of the alert-fatigue balancing loop B1 means that any intervention that amplifies awareness without filtering necessarily degrades over time. Verification is therefore not a design preference but a structural condition for sustainability. This is the most consequential insight of the analysis for the next phase.

These four insights, together with the assumption-validation plan of Table X and the unintended-consequence analysis of Section VII, define the agenda for Workshop No. 2, in which the system architecture, verification logic, and data model will be specified as concrete responses to the system-level findings established here.

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TABLE XII: Operational Gap Analysis — Current vs. Target State of the Campus Safety System

Dimension	**Current state**	**Target state**	**Gap**
Incident response time	~ 11.4 min (avg.)	< 5 min	-6.4 min
Community channel adoption	$\sim 12\%$	$\geq 60\%$	-48 pp
Incident record completeness	$\sim 41\%$ of events logged	$\geq 85\%$	-44 pp
Alert delivery success rate	$\sim 87\%$ (SMS/call)	$\geq 99\%$	-12 pp